Research — Flatpak & Flathub

Nov 4, 2026 · @Tim

Summary

Target `org.gnome.Platform` 51 (GNOME 51, September 2026, on freedesktop-sdk 26.08), launch store games through the OpenURI portal, and keep host spawning optional; that combination gives the smallest permission set a moderator will review. Every Proton-capable launcher on Flathub (Heroic, Lutris, Bottles, Faugus) runs Wine inside its own sandbox with `--device=all`, `--allow=multiarch,devel,per-app-dev-shm` and the 32-bit Compat/GL32 extensions. Cartridges, the only GNOME-native library launcher, instead used `flatpak-spawn --host` with `--filesystem=host`, and is now end-of-life on Flathub.

Hard constraint: Flathub's generative-AI policy. The Flathub manifest must not contain AI-generated or AI-assisted content; AI-generated app code, docs or packaging must be disclosed with its rough extent; and AI must not open or automate Flathub PRs or write their descriptions or replies. Exception requests to the linter must not be handled with LLMs either. The May 2026 ban was revised to this disclosure rule in autumn 2026 (sources give early September and 2026-10-08). This doc describes permissions in prose for that reason; the manifest itself has to be written by a person.

App ID, MetaInfo and submission requirements

Pick the app ID once (renames are mostly refused), ship complete MetaInfo, build a real commit history first, and submit through the `new-pr` branch of `flathub/flathub`.

App ID (Requirements)

- Reverse-DNS, 3 to 5 components, characters `[A-Za-z0-9_]`, a dash only in the last component; must match the MetaInfo `<id>`; must not end in `.desktop`, `.app` or `.linux`.
- GitHub-hosted apps use `io.github.<user>.<App>` (at least four components; `com.github.` is reserved). Dashes in the username become underscores, so a repo under the GitHub account Tim-van-Oudheusden would give `io.github.Tim_van_Oudheusden.<AppName>`.
- Verification for `io.github.*` is done by logging in to Flathub with the owning GitHub account.

MetaInfo, mandatory (guidelines)

- `<id>`, name, summary, description (only p, ol, ul, li, em, code).
- `<developer id="…"><name>`, `<launchable type="desktop-id">`.
- At least one screenshot, linked to a tag or commit, one marked default, with captions.
- `<releases>` with no future dates, correctly ordered; `<content_rating type="oars-1.1">`.
- Homepage URL (vcs-browser recommended), `metadata_license`, SPDX `project_license`, complete English text, SVG icon or PNG of at least 256 px.
- Recommended: light and dark brand colours, and `<supports><control>gamepad</control>` for the fullscreen mode (never under `requires`).
- Linter errors that are never excepted include a missing launchable, missing developer name, missing screenshots, desktop-file icon errors and `flatpak run` in a desktop file's Exec.

Quality guidelines (needed to be featured)

- Name ideally 15 characters or fewer; summary 10–25 characters, imperative, without the app name, "app", "tool" or a final period.
- Icon without baked shadows, not filling the canvas, readable on light and dark.
- 3–6 windowed screenshots at up to 1000×700 (2000×1400 HiDPI), default settings, not only dark style. A windowed library app should use windowed shots, not fullscreen-mode shots.

Inclusion-policy risks

- Thin wrappers or launchers around other tools are not accepted: the app must add substantial functionality of its own.
- Apps relying on host components for core functionality are not accepted except case by case, which matters if launching depends on host spawn.
- Very new apps with insufficient development history are generally refused; expect to need sustained commits and tagged releases.
- The Wine rule targets Windows software packaged with Wine, not native launchers (Lutris, Bottles, Heroic and Faugus are accepted; this is an interpretation).

Submission (process)

1. Build with `flatpak run --command=flathub-build org.flatpak.Builder --install <manifest>`.

2. Lint the manifest and the built repo with `flatpak-builder-lint` from the same tool.
3. Fork `flathub/flathub` (all branches), branch from `new-pr`, open a PR titled "Add <app-id>" against `new-pr`, explain every permission, comment `bot, build`.
4. Volunteers review with no deadline; after merge, accept the invite to `flathub/<app-id>` within a week with GitHub 2FA on. Builds publish in 1–2 hours.
5. Later builds that add permissions are held for moderation.

Runtime, SDK extensions, manifest and CI

Build on GNOME 51 with the matching 26.08 SDK extensions, vendor all dependencies at build time, and run flatpak-builder in GitHub Actions on the `gnome-51` image.

Item	Detail
Runtime	`org.gnome.Platform` / `Sdk` 51; final release followed the RC of 2026-09-07; based on freedesktop-sdk 26.08; drops gcr-3 and libhandy (irrelevant for libadwaita)
Support window	GNOME runtimes are usually supported about a year, until the next-but-one release; GNOME 50 runs to April 2027, so 51 likely to about September 2027 (unconfirmed). Freedesktop 26.08 runs about two years
Platform contents	libadwaita, libportal, PyGObject and libmanette 0.2 are in the GNOME 51 platform
32-bit	The GNOME runtime dropped its own compat extension in October 2025; launchers add `org.freedesktop.Platform.Compat.i386` 26.08 and `org.freedesktop.Platform.GL32` through `add-extensions` (only needed if Wine runs in the sandbox)
Rust	`org.freedesktop.Sdk.Extension.rust-stable` (branch 26.08), append `/usr/lib/sdk/rust-stable/bin` to PATH, `CARGO_NET_OFFLINE=true`; generate `cargo-sources.json` with `flatpak-cargo-generator.py` or ship a `cargo vendor` tarball

Item	Detail
Python	`flatpak-pip-generator --runtime=org.gnome.Sdk//51` from requirements or pyproject; builds have no network
Manifest basics	id, runtime, runtime-version, sdk, command, finish-args, sdk-extensions, add-extensions, cleanup, modules (meson, git sources pinned by tag and commit); licences to `$FLATPAK_DEST/share/licenses/$FLATPAK_ID`
CI	`flatpak/flatpak-github-actions/flatpak-builder@v6` in container `ghcr.io/flathub-infra/flatpak-github-actions:gnome-51` with `--privileged`; inputs bundle, manifest-path, cache-key, arch, run-tests; aarch64 via QEMU or `ubuntu-24.04-arm` runners

Tools: flatpak-builder-tools (cargo, pip), Flatpak manifests, Flathub runtimes.

Permissions a game launcher needs

A library that reads other launchers read-only and launches through portals needs only display, network and a list of `:ro` paths; running Proton in the sandbox or spawning on the host is what makes the permission set heavy. Sources: Flatpak sandbox permissions, Flathub linter, linter finish-args checks.

Permission	Why we would need it	Flathub stance	Alternative
`--socket=wayland`, `--socket=fallback-x11`, `--share=ipc`, `--device=dri`	GTK4 UI	Recommended set; `fallback-x11` without wayland, or with `x11`, is an error	If games run inside the sandbox, `x11` + `wayland` (a warning only)
`--share=network`	Cover art, SteamGridDB, Proton downloads	No issue	–
`~/.steam:ro`, `xdg-data/Steam:ro`	Native Steam libraries	Subpaths fine; whole `~/.local`, `~/.config`, `xdg-data` are errors	–
`~/.var/app/com.valvesoftware.Steam/data/Steam:ro`	Flatpak Steam	No rule; `home` does not include `~/.var/app`, so list each path	–

Permission	Why we would need it	Flathub stance	Alternative
`~/.config/heroic:ro`, `~/.var/app/com.heroicgameslauncher.hgl/config/heroic:ro`	Heroic	Fine	–
`~/.local/share/lutris:ro`, `~/.var/app/net.lutris.Lutris/data/lutris:ro`	Lutris	Fine	–
`~/.var/app/com.usebottles.bottles/data/bottles:ro`, Faugus data dirs `:ro`	Bottles, Faugus	Fine	–
`xdg-data/applications:ro`, `host-os:ro`, Flatpak `exports:ro` dirs	Desktop entries and Flatpak games	No rule, moderators review	–
`--filesystem=home` or `host`	Manual games, prefixes	Not a linter error (both together never granted), but heavy	File chooser portal plus `~/Games:create`
`--talk-name=org.freedesktop.Flatpak`	`flatpak-spawn --host`	Warning only, but a full sandbox escape judged by moderators	OpenURI portal; Proton in sandbox
`--device=input`	Gamepads via evdev (libmanette)	Error unless `--require-version>=1.16.0`, or paired as `--device-if=all:!has-input-device` + `--device=input`	–
`--device=all`	hidraw (Steam Deck), uinput, games in sandbox	Allowed; used by all comparable launchers	`input` if only the UI reads gamepads
`--allow=multiarch,devel,per-app-dev-shm`, `--socket=pulseaudio`, `xdg-data/umu:create`	umu, pressure-vessel and Wine inside the sandbox	Precedent in Heroic, Lutris, Bottles, Faugus	Not needed if games launch elsewhere
GameMode	Performance	`--talk-name=org.freedesktop.portal.*` is an error, never granted	GameMode portal via libgamemode 1.4+
Background, OpenURI, Notification, Inhibit portals	Autostart, store URIs, notifications, no-sleep	No static permission needed	–

Linter exceptions are PRs to `flathub/flatpak-builder-lint` `exceptions.json`, case by case and meant to be temporary. Passing the linter does not approve the permissions; moderators decide separately.

Launching games from the sandbox

Recommended: OpenURI for every store game, in-sandbox umu for standalone Windows games, and host spawning only as an opt-in the user grants with `flatpak override`.

Option	How	Strengths	Costs
A. Host spawn	`flatpak-spawn --host …` via `org.freedesktop.Flatpak`	Runs anything on the host: native Steam, Lutris, Heroic, `flatpak run <id>`, host `umu-run`, `.desktop` Exec lines; proven by Cartridges	Sandbox escape and moderator scrutiny; "relies on host" policy risk; exit status only for the direct child, so URI launches return at once
B. OpenURI portal	In-sandbox `xdg-open steam://rungameid/<id>`, `heroic://launch?…`, `lutris:rungameid/<id>`	No static permission; works for native and Flatpak stores (Heroic and Lutris register their schemes)	Fire and forget: no PID or exit; may show a chooser with no default; Faugus and Bottles schemes unconfirmed
C. Other Flatpaks	`flatpak-spawn --host flatpak run com.heroicgameslauncher.hgl …`	Direct control of flags	Same costs as A; B usually avoids it
D. umu in the sandbox	Bundle umu-launcher (Bottles ships it), Compat.i386, GL32, multiarch, devel, per-app-dev-shm, `--device=all`, x11, pulseaudio, `xdg-data/umu:create`, game dir access	Self-contained; same model as Lutris, Bottles, Heroic, Faugus; game PIDs visible, so playtime is exact	Heavy manifest and broad permissions; pressure-vessel needs a recent Flatpak and unprivileged user namespaces
E. Sub-sandbox spawn	`flatpak-spawn` without `--host`	No escape; PID sharing flags	Only spawns copies of our own sandbox; cannot reach host Steam

Playtime for host-launched games. The sandbox has its own PID namespace, so `/proc` cannot see host games. With A we can wait on the direct child (native binaries, host `umu-run`); URI launches return immediately, so playtime must come from the store's own data

(Heroic `timestamp.json`; Steam's location unconfirmed) or host polling (unconfirmed approach).

References: flatpak-spawn(1), OpenURI portal, flatpak-xdg-utils, umu Flatpak packaging, Flathub forum on Flatpak inside Flatpak.

Comparable launchers' finish-args

Every accepted Proton launcher grants itself broad device and multiarch access; only Cartridges took the host-spawn route, and it is now end-of-life. Taken from Flathub master manifests on 2026-11-04.

App	Runtime	Key finish-args
Cartridges v2.13.1, end-of-life	GNOME 49	network, ipc, wayland, fallback-x11, dri; talk `org.freedesktop.Flatpak`; `--filesystem=host`; read-only paths for Flatpak Steam, Lutris, Heroic, Bottles, itch, RetroArch and Flatpak exports
Heroic	Freedesktop 26.08 + Electron BaseApp; Compat.i386, GL32	allow devel, multiarch, bluetooth, per-app-dev-shm; device=all; read-write Steam, Lutris and applications dirs; `~/Games/Heroic:create`; `/run/udev:ro`; `xdg-data/umu:create`; x11, wayland, pulseaudio
Lutris	GNOME 51 + org.winehq.Wine 26.08; Compat.i386, GL32, gamescope, MangoHud	x11, wayland, pulseaudio, device=all, allow multiarch, devel, per-app-dev-shm; talk `org.freedesktop.Flatpak`; home, `~/Games`, Flatpak Steam `:ro`, `xdg-data/umu:create`; builds gamemode
Bottles	GNOME 50 + Wine 25.08; Compat.i386, GL32	allow devel, multiarch, per-app-dev-shm; device=all; x11, wayland, pulseaudio; `xdg-data/umu:create`, `~/Games/umu:create`; no home access; bundles umu-launcher 1.4.4

App	Runtime	Key finish-args
Faugus v2.5.7	GNOME 51; Compat.i386, GL32	allow devel, bluetooth, multiarch, per-app-dev-shm; device=all; home, `/mnt`, `/run/media`; own `com.steampowered.*`; x11, wayland, pulseaudio
Steam	Freedesktop 26.08	device=all, allow multiarch, devel, bluetooth, per-app-dev-shm; `/run/udev:ro`; own `com.steampowered.*`; require Flatpak 1.12.0

umu-launcher itself does not appear to be on Flathub; upstream's Flatpak manifest targets the outdated freedesktop 23.08 runtime.

Gamepad and device access

For gamepad navigation of our own UI, `--device=input` (paired with `--device-if=all:!has-input-device`) is enough for ordinary controllers; Steam Deck controls need hidraw, which only `--device=all` exposes.

- `--device=input` (Flatpak 1.15.6, November 2023) bind-mounts only `/dev/input`; raw and virtual input (`/dev/hidraw*`, `/dev/uinput`) still need `--device=all` (Flatpak NEWS, flatpak-run.c).
- libmanette always uses its file backend inside Flatpak (since 0.2.7), watching `/dev/input/event*` and `/dev/hidraw*` without udev; Steam Deck support (0.2.11+, hidapi-hidraw) needs hidraw (manette-monitor.c, NEWS).
- SDL disables udev in containers and watches `/dev/input` with inotify; its HIDAPI drivers would need hidraw (unconfirmed in Flatpak).
- Device permissions still come from host udev rules (`steam-devices`); Heroic and Steam add `/run/udev:ro` for Deck navigation (Steam Flatpak wiki).

Open questions

- Exact GNOME 51 end-of-life date.
- Would moderators accept `org.freedesktop.Flatpak` plus broad file access for a new launcher today, given the host-dependency and thin-wrapper rules?

- Do Bottles and Faugus expose URI schemes for external launching, and where does Steam store playtime?
- Does the freedesktop runtime ship `gamemoderun`?
- Where exactly does the AI disclosure go in a submission?

Sources

- Flathub docs: Requirements, Linter, MetaInfo guidelines, Quality guidelines, Submission, Runtimes, Verification, Maintenance
- Flatpak docs: Sandbox permissions, Manifests, Available runtimes, flatpak-spawn(1), NEWS, flatpak-run.c, portal.Flatpak D-Bus API, Flatpak D-Bus API, flatpak-xdg-utils
- Portals: OpenURI, GameMode, Background
- Linter: exceptions.json, finish_args.py
- Build tooling: flatpak-builder-tools, flatpak-github-actions, actions-images README
- Comparable manifests: Cartridges and its flathub.json, Heroic, Lutris, Bottles and its umu module, Faugus, Steam, Steam Flatpak wiki
- umu: README, Flatpak packaging, umu_run.py
- Gamepads: libmanette manette-monitor.c, libmanette NEWS, SDL Linux joystick, Flathub forum: gamepads
- GNOME runtime: gnome-51 sdk-platform.bst, GNOME 51 RC, GNOME 51 drops gcr-3 and libhandy, 32-bit extension removed (Linuxiac), GNOME 48 runtime EOL
- AI policy coverage: Linuxiac, GamingOnLinux
- AppStream metadata spec